Requirement Analysis Phase

Technology Stack Template

Phase 2: Requirement Analysis

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Technology Stack Overview

The shopEZ platform is built on the MERN stack (MongoDB, Express.js, React.js, Node.js), supplemented by carefully selected supporting libraries to ensure security, performance, and developer productivity.

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2. Layer-by-Layer Technology Table

Layer	Technology	Version	Justification
Frontend Framework	React.js	19.x	Component-driven SPA with virtual DOM for blazing-fast UI updates without full page reloads.

Build Tooling	Vite	8.x	Lightning-fast HMR development server with optimized production builds and native ES module support.
Styling	Tailwind CSS	3.x	Utility-first responsive design with custom HSL color palettes, eliminating unused CSS at build.
Client-Side Routing	React Router DOM	6.x	Declarative route hierarchy with protected route guards and nested layout support.
HTTP Client	Axios	1.7.x	Promise-based HTTP client with request/response interceptors for automatic auth header injection.
Analytics Charts	Chart.js + react-chartjs-2	4.x	Canvas-rendered bar, doughnut, and line charts for Admin Dashboard analytics visualizations.
Icon Library	Lucide React	0.395.x	Consistent, accessible SVG icon library with tree-shaking for optimized bundle sizes.

Backend Runtime	Node.js	18+	Server-side JavaScript enabling full-stack JS development with non-blocking event-driven I/O.
Web Framework	Express.js	4.x	Minimal, unopinionated RESTful API framework with rich middleware ecosystem.
Database	MongoDB + Mongoose	8.x	NoSQL document store with schema validation via Mongoose ODM, ideal for flexible product catalogs.
Authentication	JWT + Bcrypt.js	Latest	Stateless token sessions with 30-day expiry + secure password hashing with salt rounds of 10.
Image Storage	Multer (Local Disk)	1.4.x	Disk-based image persistence with Express static file serving from /backend/uploads directory.
Dev Fallback DB	mongodb-memory-server	11.x	In-memory MongoDB for development without external DB configuration or cloud dependency.

Dev Server Utility	Nodemon	3.x	Auto-restart server on file changes, eliminating manual restart cycles during development.

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3. Technology Selection Rationale

> **Why MERN?** The MERN stack enables full JavaScript across the entire application — from database schemas (Mongoose) to API logic (Express) to UI components (React) — dramatically reducing context-switching overhead and enabling shared validation logic between frontend and backend.

> **Why Vite over Create-React-App?** Vite uses native ES modules in development, delivering sub-second HMR even in large codebases. CRA's Webpack-based setup becomes prohibitively slow at scale. Vite's Rollup production bundler also generates significantly smaller output bundles.

> **Why Local Multer over Cloudinary?** For a development and demonstration project, local Multer disk storage eliminates the need for external API credentials and CDN configuration. The architecture is designed for simple migration to Cloudinary or AWS S3 by swapping the Multer storage engine.

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4. Architecture Stack Diagram

Frontend Stack	Backend Stack	Data Layer
React.js 19.x Vite 8.x Tailwind CSS 3.x React Router DOM 6.x Axios 1.7.x Chart.js 4.x Lucide React	Node.js 18+ Express.js 4.x JWT (jsonwebtoken) Bcrypt.js Multer 1.4.x Nodemon 3.x	MongoDB 8.x Mongoose ODM mongodb-memory-server 11.x